

Agentic Vol Regime App

Decision-support frontend on top of agentic_harness.

Daily Belief Report Historical Belief Reports **Backtester** IBKR Snapshot Agent Resume Daily Review

Backtester

Deterministic historical replay with strict no-lookahead constraints. This mode is fully offline and does not call IBKR.

Replay Config Path

C:\Users\joyba\PycharmProjects\langchain_langgraph\agentic_vol_regime_app\configs\backtest\hmm_replay.yaml

Feature Store Builder (IBKR)

Symbol	History Days	IBKR Host	IBKR Port
SPY	1512 - +	127.0.0.1	4001 - +
Client ID	Market Data Type	Exchange	Index Exchange
73 - +	1 - +	SMART	CBOE

Feature Store As-of Date (optional historical cut) ?

2026/06/12

Build / Refresh Feature Store

Feature store: C:\Users\joyba\PycharmProjects\langchain_langgraph\agentic_vol_regime_app\agentic_vol_regime_app\data\processed\features_daily.parquet

Replay Mode

☒ Date Range ☐ Single Date

Models

hmm_v3_core_plu... x

Horizons (trading days)

1 x 2 x 3 x

Start Date	End Date
2026/04/01	2026/06/05

Run Backtest

Report: reports\backtests\hmm_replay\run_20260613T014830Z\replay_report.md

Summary CSV: reports\backtests\hmm_replay\run_20260613T014830Z\summary_metrics.csv

Predictions JSONL: data\backtests\hmm_replay\run_20260613T014830Z\prediction_records.jsonl

Outcomes JSONL: data\backtests\hmm_replay\run_20260613T014830Z\outcome_records.jsonl

Scored JSONL: data\backtests\hmm_replay\run_20260613T014830Z\scored_records.jsonl

Summary Metrics

model_name	horizon	accuracy	brier_vol_expansion	brier_vix_spike	avg_lead_quality
hmm_v3_core_plus_sector_geometry	1	0.3125	0.4632	0.1508	
hmm_v3_core_plus_sector_geometry	2	0.2917	0.5101	0.1602	
hmm_v3_core_plus_sector_geometry	3	0.3125	0.4479	0.1423	

Replay Report

HMM Replay Backtest Report

Overall Model Comparison

model_name	horizon	accuracy	brier_vol_expansion	brier_vix_spike	avg_lead_quality	notes
hmm_v3_core_plus_sector_geometry	1	0.3125	0.463165	0.1508	0.479167	
hmm_v3_core_plus_sector_geometry	2	0.291667	0.510124	0.160224	0.354167	
hmm_v3_core_plus_sector_geometry	3	0.3125	0.447855	0.142286	0.395833	

Recent As-of Date Comparison

As Of	Model	Predicted State	T+1 Outcome	T+2 Outcome	T+3 Outcome
2026-04-27	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	MID_VOL_CHOP	MID_VOL_CHOP
2026-04-28	hmm_v3_core_plus_sector_geometry	MID_VOL_CHOP	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	STABLE_LOW_VOL_TREND
2026-04-29	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-04-30	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-05-01	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION
2026-05-04	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	STABLE_LOW_VOL_TREND
2026-05-05	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-05-06	hmm_v3_core_plus_sector_geometry	MID_VOL_CHOP	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-05-07	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION
2026-05-08	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION
2026-05-11	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	STABLE_LOW_VOL_TREND
2026-05-12	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND

As Of	Model	Predicted State	T+1 Outcome	T+2 Outcome	T+3 Outcome
2026-05-13	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	VOL_EXPANSION
2026-05-14	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-15	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-18	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	STABLE_LOW_VOL_TREND	MID_VOL_CHOP
2026-05-19	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-20	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-21	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-22	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-25	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-05-26	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-05-27	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND
2026-05-28	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	MID_VOL_CHOP	STABLE_LOW_VOL_TREND
2026-05-29	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	MID_VOL_CHOP	STABLE_LOW_VOL_TREND	VOL_EXPANSION
2026-06-01	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION
2026-06-02	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	HIGH_VOL_RISK_ON
2026-06-03	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	STABLE_LOW_VOL_TREND	HIGH_VOL_RISK_OFF	VOL_EXPANSION
2026-06-04	hmm_v3_core_plus_sector_geometry	STABLE_LOW_VOL_TREND	HIGH_VOL_RISK_OFF	HIGH_VOL_RISK_OFF	HIGH_VOL_RISK_ON
2026-06-05	hmm_v3_core_plus_sector_geometry	VOL_EXPANSION_TRANSITION	VOL_EXPANSION_TRANSITION	MID_VOL_CHOP	VOL_EXPANSION

Model Disagreement Cases

No rows

HMM v3 Special Section

Track whether geometry features improved false vol-expansion avoidance and mid-vol chop detection.

Diagnostics

as_of_date	model_name	converged	training_row_count	training_end_date	warnings
2026-04-01	hmm_v3_core_plus_sector_geometry	True	756	2026-04-01	
2026-04-02	hmm_v3_core_plus_sector_geometry	True	756	2026-04-02	
2026-04-03	hmm_v3_core_plus_sector_geometry	True	756	2026-04-03	
2026-04-06	hmm_v3_core_plus_sector_geometry	True	756	2026-04-06	
2026-04-07	hmm_v3_core_plus_sector_geometry	True	756	2026-04-07	
2026-04-08	hmm_v3_core_plus_sector_geometry	True	756	2026-04-08	
2026-04-09	hmm_v3_core_plus_sector_geometry	True	756	2026-04-09	
2026-04-10	hmm_v3_core_plus_sector_geometry	True	756	2026-04-10	
2026-04-13	hmm_v3_core_plus_sector_geometry	True	756	2026-04-13	
2026-04-14	hmm_v3_core_plus_sector_geometry	True	756	2026-04-14	
2026-04-15	hmm_v3_core_plus_sector_geometry	True	756	2026-04-15	
2026-04-16	hmm_v3_core_plus_sector_geometry	True	756	2026-04-16	
2026-04-17	hmm_v3_core_plus_sector_geometry	True	756	2026-04-17	
2026-04-20	hmm_v3_core_plus_sector_geometry	True	756	2026-04-20	
2026-04-21	hmm_v3_core_plus_sector_geometry	True	756	2026-04-21	
2026-04-22	hmm_v3_core_plus_sector_geometry	True	756	2026-04-22	
2026-04-23	hmm_v3_core_plus_sector_geometry	True	756	2026-04-23	
2026-04-24	hmm_v3_core_plus_sector_geometry	True	756	2026-04-24	
2026-04-27	hmm_v3_core_plus_sector_geometry	True	756	2026-04-27	
2026-04-28	hmm_v3_core_plus_sector_geometry	True	756	2026-04-28	
2026-04-29	hmm_v3_core_plus_sector_geometry	True	756	2026-04-29	
2026-04-30	hmm_v3_core_plus_sector_geometry	True	756	2026-04-30	
2026-05-01	hmm_v3_core_plus_sector_geometry	True	756	2026-05-01	
2026-05-04	hmm_v3_core_plus_sector_geometry	True	756	2026-05-04	
2026-05-05	hmm_v3_core_plus_sector_geometry	True	756	2026-05-05	
2026-05-06	hmm_v3_core_plus_sector_geometry	True	756	2026-05-06	
2026-05-07	hmm_v3_core_plus_sector_geometry	True	756	2026-05-07	
2026-05-08	hmm_v3_core_plus_sector_geometry	True	756	2026-05-08	
2026-05-11	hmm_v3_core_plus_sector_geometry	True	756	2026-05-11	
2026-05-12	hmm_v3_core_plus_sector_geometry	True	756	2026-05-12	
2026-05-13	hmm_v3_core_plus_sector_geometry	True	756	2026-05-13	
2026-05-14	hmm_v3_core_plus_sector_geometry	True	756	2026-05-14	
2026-05-15	hmm_v3_core_plus_sector_geometry	True	756	2026-05-15	
2026-05-18	hmm_v3_core_plus_sector_geometry	True	756	2026-05-18	
2026-05-19	hmm_v3_core_plus_sector_geometry	True	756	2026-05-19	

as_of_date	model_name	converged	training_row_count	training_end_date	warnings
2026-05-20	hmm_v3_core_plus_sector_geometry	True	756	2026-05-20	
2026-05-21	hmm_v3_core_plus_sector_geometry	True	756	2026-05-21	
2026-05-22	hmm_v3_core_plus_sector_geometry	True	756	2026-05-22	
2026-05-25	hmm_v3_core_plus_sector_geometry	True	756	2026-05-25	
2026-05-26	hmm_v3_core_plus_sector_geometry	True	756	2026-05-26	
2026-05-27	hmm_v3_core_plus_sector_geometry	True	756	2026-05-27	
2026-05-28	hmm_v3_core_plus_sector_geometry	True	756	2026-05-28	
2026-05-29	hmm_v3_core_plus_sector_geometry	True	756	2026-05-29	
2026-06-01	hmm_v3_core_plus_sector_geometry	True	756	2026-06-01	
2026-06-02	hmm_v3_core_plus_sector_geometry	True	756	2026-06-02	
2026-06-03	hmm_v3_core_plus_sector_geometry	True	756	2026-06-03	
2026-06-04	hmm_v3_core_plus_sector_geometry	True	756	2026-06-04	
2026-06-05	hmm_v3_core_plus_sector_geometry	True	756	2026-06-05	

> Backtest Artifacts